



BAULKHAM HILLS HIGH SCHOOL

TRIAL 2013
YEAR 12 TASK 4

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 10 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks)
Questions 1-10

- Attempt Question 1-10

Section II – Pages 5-9 (60 marks)

- Attempt questions 11-14

Table of Standard Integrals is on page 10

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. The value of $\lim_{x \rightarrow 0} \frac{\sin 7x}{3x}$ is

(A) $2\frac{1}{3}$

(B) $\frac{3}{7}$

(C) 0

(D) 1

2. What is the acute angle, to the nearest degree, between the lines $y = 7 - 4x$ and $2x - 3y - 6 = 0$.

(A) 26°

(B) 48°

(C) 70°

(D) 75°

3. Let α, β and γ be the roots of $2x^3 + x^2 - 4x + 9 = 0$.

What is the value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$

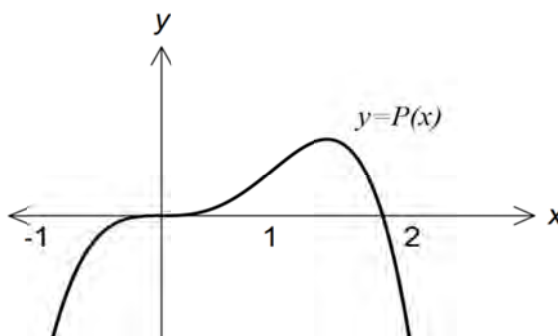
(A) $-\frac{1}{9}$

(B) $-\frac{1}{2}$

(C) $\frac{1}{9}$

(D) $\frac{1}{2}$

4. Which of the following could be the polynomial $y = P(x)$



- (A) $P(x) = x^3(2 - x)$
 (B) $P(x) = x^2(2 - x)^2$
 (C) $P(x) = x^3(x - 2)$
 (D) $P(x) = -x^3(x + 2)$

5. The coordinates of the points A and B are $(-1, 1)$ and $(3, -1)$ respectively. Find the coordinates of P that divides the interval AB externally in the ratio 1:3.

- (A) $(2, -\frac{1}{2})$
 (B) $(-1\frac{1}{2}, \frac{1}{2})$
 (C) $(-3, 2)$
 (D) $(4, -2)$

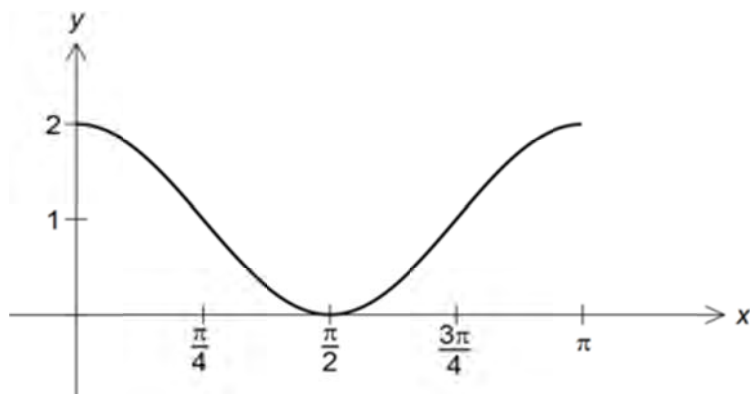
6. The number of different arrangements of the letters of the word **REGISTER** which begin and end with the letter R is:

- (A) $\frac{6!}{(2!)^2}$
 (B) $\frac{8!}{(2!)}$
 (C) $\frac{6!}{2!}$
 (D) $\frac{8!}{2!2!}$

7. The inverse function of $g(x)$ where $g(x) = \sqrt{3x - 8}$ is

- (A) $g^{-1}(x) = \frac{x^2 + 8}{3}$
 (B) $g^{-1}(x) = (3x - 8)^2$
 (C) $g^{-1}(x) = \sqrt{\frac{x}{3}} + 8$
 (D) $g^{-1}(x) = \frac{x^2 - 8}{3}$

8. A possible equation for the curve drawn is:-



- (A) $y = -1 + \cos 2x$
(B) $y = 1 + \cos 2x$
(C) $y = \cos(2x + 1)$
(D) $y = 1 - \cos 2x$
9. The middle term in the expansion $(2x - 4)^4$ is

- (A) 81
(B) $216x^2$
(C) $384x^2$
(D) $-96x^3$

10. If $y = \sin^{-1}(2x)$, then $\frac{d^2y}{dx^2}$ equals

- (A) $\frac{-8}{\sqrt{(1-4x^2)^3}}$
(B) $\frac{4x}{\sqrt{1-4x^2}}$
(C) $\frac{-8x}{\sqrt{(1-4x^2)^3}}$
(D) $\frac{8x}{\sqrt{(1-4x^2)^3}}$

End of Section 1

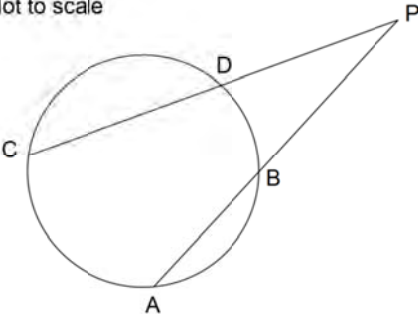
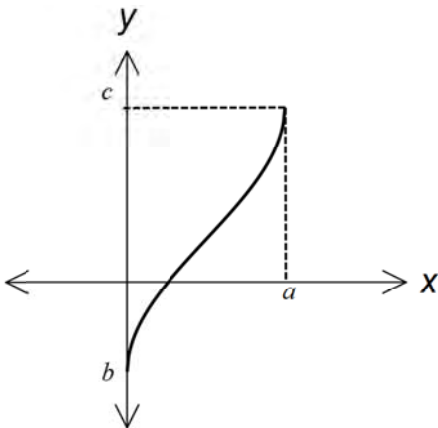
Section II – Extended Response

Attempt questions 11-16.

Answer each question on a SEPARATE PAGE. Clearly indicate question number.

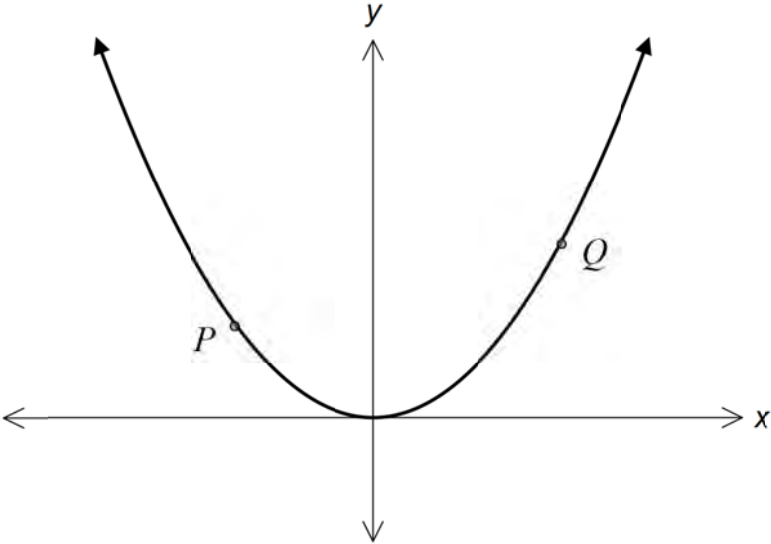
Each piece of paper must show your number.

All necessary working should be shown in every question.

Question 11 (15 marks)		Marks
a)	Solve for x $\frac{2x+1}{1-x} \geq 1$	3
b)	It is given that $y = 2 \cos 3 \left(x - \frac{\pi}{3} \right), 0 \leq x \leq 2\pi$ Find (i) the amplitude. (ii) the period.	1 1
c)	Not to scale  Two chords CD and AB of a circle are extended to meet at P. If $AB = 7\text{cm}, BP = 3\text{cm}, DP = 5\text{cm}$ find CP.	1
d)	The graph of $y = 1 + 2 \sin^{-1}(2x - 1)$ is shown.  Determine the values of a, b, and c	2
e)	Form the cartesian equation by eliminating the parameter, θ, from these parametric equations. $x = 3 \tan \theta \quad y = 2 \sec \theta$	2
f)	Solve $\sin 2\theta + \cos \theta = 0$ for $0 \leq \theta \leq 2\pi$.	3
g)	The radius of a spherical balloon is increasing at the rate of 2cm/sec . Find the rate at which the volume of the balloon is increasing when the radius is 10cm (in terms of π).	2
End of Question 11		

Question 12 (15 marks)		Marks
a)	Evaluate $\int_0^{\frac{\pi}{6}} 2 \sin^2 x \, dx$	2
b)	By using the substitution $u = \sqrt{x}$, evaluate $\int_1^4 \frac{dx}{x + \sqrt{x}}$	3
c)	Two parallel tangents to a circle, centre O , are cut by a third tangent at P and Q . (i) Copy or trace the diagram into your solution booklet. (ii) Prove $\triangle MOQ \equiv \triangle SOQ$ (iii) Prove that $\angle POQ = 90^\circ$.	2 2
d)	(i) Express $x = 2 \cos 3t - 5 \sin 3t$ in the form $x = R \cos(3t + \alpha)$, where t is in radians.	2
	(ii) A particle moves in a straight line and its position at time t is given by $x = 2 \cos 3t - 5 \sin 3t$. Show that the particle is moving in simple harmonic motion.	1
e)	The coefficients of x^2 and x^{-1} in the expansion of $\left(ax - \frac{b}{x^2}\right)^5$ are the same. Show that $a + 2b = 0$, where a and b are positive integers.	3
End of Question 12		

Question 13 (15 marks)		
a)	In how many ways can a committee of 5 people be formed from a group of 9 people, if 2 people, Harry and Archie, refuse to serve together on the same committee.	2
b)	(i) Show by a sketch, without using calculus, that the equation $e^{2x} + 4x - 5 = 0$ has only one root.	1
	(ii) Show that this root lies between 0 and 1.	1
	(iii) By taking $x = 0.5$ as a first approximation, use one application of Newton's method to find a better approximation of this root to two decimal places.	2
c)	A particle moves in a straight line with velocity v m/sec and acceleration given by $\ddot{x} = 2e^x$, where x is the displacement from O . The initial velocity is -2 m/sec at the origin.	
	(i) Prove that $v^2 = 4e^x$.	2
	(ii) Hence find the displacement in terms of t .	3
d)	When a body falls, the rate of change of its velocity, v , is given by $\frac{dv}{dt} = -k(v - 500)$ where k is a constant.	
	(i) Show that $v = 500 - 500e^{-kt}$ is a possible solution to this equation.	1
	(ii) The velocity after 5 seconds is 21m/sec, find the value of k to 3 significant figures.	1
	(iii) Find the velocity after 20 seconds.	1
	(iv) Explain the effect on the velocity as t becomes large.	1
End of Question 13		

Question 14 (15 marks)	Marks
a) A particle is projected with an initial velocity of 60m/sec at an angle of 45° to the horizontal. The equations of motion are: $x = 30\sqrt{2}t \quad \text{and} \quad y = 30\sqrt{2}t - 5t^2 \quad (\text{Do not prove these})$ (i) How high is the particle after it has travelled 120m horizontally? 2 (ii) What is the horizontal range of the particle? 1	
b) Two points P and Q move on the parabola $x^2 = 4ay$ so that their x values differ by a.  (i) If P has coordinates (x_1, y_1), show that the midpoint, C, of PQ is $\left(\frac{2x_1 + a}{2}, \frac{2x_1^2 + 2ax_1 + a^2}{8a} \right)$ (ii) Show that the locus of C is another parabola. 2	
Question 14 continues on page 6	

Question 14 (continued)

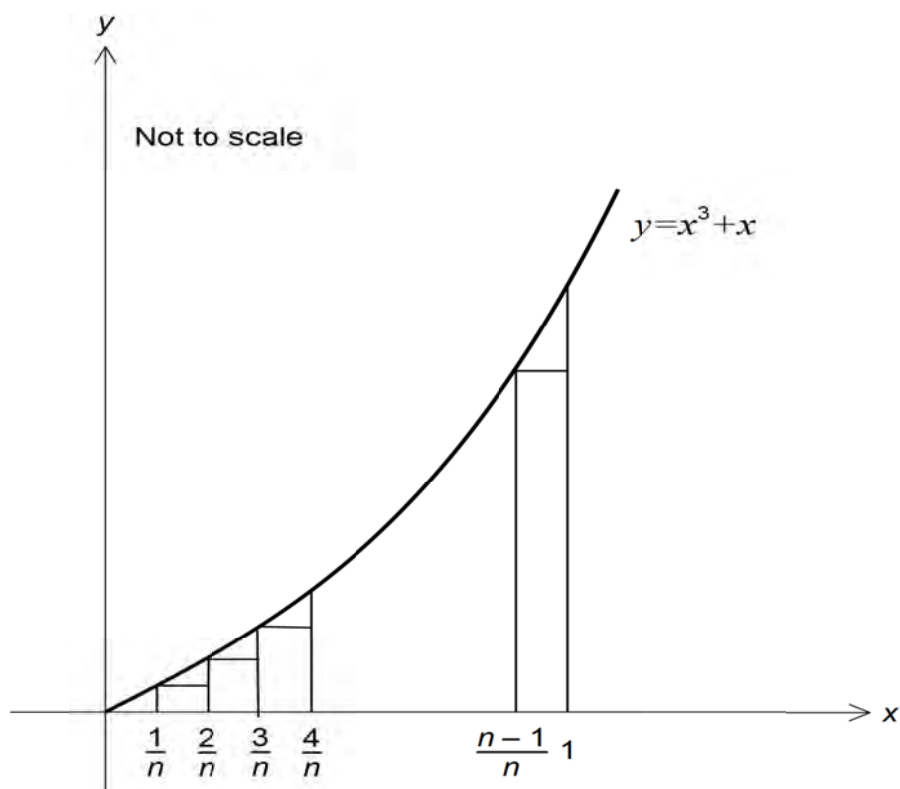
- c) (i) Prove by mathematical induction

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$$

3

- (ii) The curve $y = x^3 + x$ is drawn and divided into n intervals between 0 and 1. The inner rectangles are completed as shown.

3



If A_n is the area of these rectangles show that

$$A_n = \frac{1}{n^4} \sum_{r=1}^{n-1} r^3 + \frac{1}{n^2} \sum_{r=1}^{n-1} r$$

- (iii) Given that the area bounded by the curve, between $x = 0$ and $x = 1$, and the x axis is $\lim_{n \rightarrow \infty} A_n$, find a value for A_n using the result in part (i).

2

End of Exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 11.

a. $\frac{2x+1}{1-x} \geq 1$

various methods

example

$$\frac{2x+1-1+x}{1-x} \geq 0$$

✓ working

$$(1-x) \times 3x \geq 0 \times (1-x)^2$$

$$(1-x)(3x) \geq 0$$

$$0 \leq x \leq 1$$

but $x \neq 1$ ✓

$$\therefore 0 \leq x < 1$$

③

b. i) amplitude = 2 ✓

ii) period = $\frac{2\pi}{3}$ ✓

②

c. $CP \times DP = AP \times BP$

$$CP \times 5 = 10 \times 3$$

$$CP = 6 \text{ cm}$$

①

d. Domain: $-1 \leq 2x-1 \leq 1$

$$0 \leq x \leq 1$$

Range: $-\frac{\pi}{2} \leq \frac{y-1}{2} \leq \frac{\pi}{2}$

$$1-\pi \leq y \leq 1+\pi$$

$$\therefore a = 1$$

$$b = 1-\pi$$

$$c = 1+\pi$$

for both

②

e. $\frac{x}{3} = \tan \theta$ $\frac{y}{2} = \sec \theta$ ✓ working

connector: $\tan^2 \theta + 1 = \sec^2 \theta$

$$\therefore \frac{x^2}{9} + 1 = \frac{y^2}{4}$$

$$\text{or } 4x^2 - 9y^2 = -36$$

②

f. $\sin 2\theta + \cos \theta = 0$ $0 \leq \theta \leq 2\pi$

$$2\sin \theta \cos \theta + \cos \theta = 0$$

$$\cos \theta (2\sin \theta + 1) = 0$$

$$\therefore \cos \theta = 0 \quad \sin \theta = -\frac{1}{2}$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2} \quad \theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore \theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

③

g. $V = \frac{4}{3}\pi r^3$ $\frac{dr}{dt} = 2 \text{ cm/sec}$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$= 4\pi r^2 \times 2$$

at $r=10$

$$\frac{dV}{dt} = 800\pi \text{ cm}^3/\text{sec}$$

②

multiple choice

1. A

6. C

2. C

7. A

3. C

8. B

4. A

9. C

5. C

10. D

Question 12

a. $\int_0^{\frac{\pi}{6}} 2\sin^2 x dx = \int_0^{\frac{\pi}{6}} (1 - \cos 2x) dx$

$$= \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{6}}$$

$$= \left(\frac{\pi}{6} - \frac{1}{2} \sin \frac{\pi}{3} - 0 \right)$$

$$= \frac{\pi}{6} - \frac{\sqrt{3}}{4} \quad \text{or} \quad \frac{2\pi - 3\sqrt{3}}{12}$$

②

b. $\int_1^4 \frac{dx}{x+\sqrt{x}}$

$$u = x^{\frac{1}{2}} \quad \text{at } x=4 \quad u=2$$

$$\frac{du}{dx} = \frac{1}{2}x^{-\frac{1}{2}} \quad x=1 \quad u=1$$

$$2u du = dx$$

$$= \int_1^2 \frac{2u du}{u^2 + u}$$

$$= \int_1^2 \frac{2 du}{u+1}$$

$$= \left[2 \ln(u+1) \right]_1^2$$

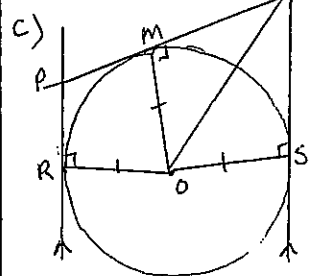
$$= 2 \ln 3 - 2 \ln 2$$

$$= 2 \ln \frac{3}{2}$$

③

Suggested solutions
and marking scale
for Trial yr12
Ext1 2013.

• variations do occur



xi) Aim: prove $\triangle MOQ \equiv \triangle SOQ$.

Method:

In $\triangle MOQ$ and $\triangle SOQ$.

$MO = OS$ (radii)

OQ is common

$\angle MDO = \angle SDO$ (radii to tangent is $\perp$)
 $= 90^\circ$

$\therefore \triangle MOQ \equiv \triangle SOQ$ (R.H.S)

② marks - complete proof

① mark - 2 reasons.

②

Quest 12 Cont.

(iii) An example

Similar to (ii) $\triangle MOP \equiv \triangle ROP$ (R.H.S)

$$\text{let } \hat{mOP} = y$$

$= \hat{ROP}$ (matching $\angle$ s in congruent $\triangle$ s)

and let $\hat{RPO} = z$

$= \hat{mPO}$ (matching $\angle$ s in congruent $\triangle$ s)

Since tangent PQ meets RP and SQ (parallel tangents)

$$\hat{RPQ} + \hat{SQM} = 180 \text{ (co-interior } \angle\text{s, } PR \parallel QS)$$

$$\therefore 2x + 2y = 180$$

$$x + y = 90 \text{ --- (1)}$$

$$\text{now } \hat{POM} = 90 - x \text{ (}\angle \text{sum of } \triangle POM)$$

$$\hat{QOM} = 90 - y \text{ (}\angle \text{sum of } \triangle QOM)$$

$$\text{then } \hat{POQ} = 90 - x + 90 - y \text{ (addition of adjacent } \angle\text{s)}$$

$$= 180 - (x + y)$$

* needed to establish

$$\text{using (1)} \quad = 180 - 90$$

$$= 90^\circ \text{ as required.}$$

ROS was a straight line!

* various methods. (1) mark connecting (ii)

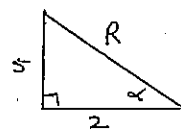
(2) marks-clear proof

$$d) x = 2 \cos 3t - 5 \sin 3t$$

$$x = R \cos 3t \cos \alpha - R \sin 3t \sin \alpha$$

$$R \cos \alpha = 2$$

$$R \sin \alpha = 5$$



$$R = \sqrt{29} \checkmark$$

$$\tan \alpha = \frac{5}{2}$$

$$\alpha = 1.19 \text{ (2dp)}$$

$$\text{or } = \tan^{-1}\left(\frac{5}{2}\right)$$

$$\therefore x = \sqrt{29} \cos(3t + 1.19) \checkmark \text{ or } \sqrt{29} \cos(3t + \tan^{-1}\frac{5}{2})$$

(2)

Question 12 Cont.

$$d) (ii) x = \sqrt{29} \cos(3t + 1.19)$$

$$\dot{x} = -3\sqrt{29} \sin(3t + 1.19)$$

$$\ddot{x} = -9\sqrt{29} \cos(3t + 1.19)$$

$$\checkmark = -9 \times x \text{ now defn of SHM is } \ddot{x} = -n^2 x$$

$\therefore$ the particle is moving in SHM with $n=3$.

(1)

$$e) \left(ax - \frac{b}{x^2}\right)^5 = (ax - bx^{-2})^5$$

$$T_{k+1} = {}^5C_k a^{5-k} b^k \text{ for } (a+b)^n$$

$$= {}^5C_k (ax)^{5-k} \cdot (-1)^k \cdot (bx^{-2})^k \checkmark$$

$$= {}^5C_k a^{5-k} \cdot (-1)^k \cdot b^k \cdot x^{5-k} \cdot x^{-2k}$$

to find k.

$$\text{for } x^2: 2 = 5 - 3k$$

$$3k = 3$$

$$k = 1$$

$$\text{Coef of } T_2 = {}^5C_1 a^4 (-b)^1$$

$$= -5a^4 b \checkmark$$

$$\text{for } x^{-1}: -1 = 5 - 3k$$

$$3k = 6$$

$$k = 2$$

$$\text{Coef of } T_3 = {}^5C_2 a^3 b^2$$

$$= 10a^3 b^2 \checkmark$$

Since coefs of x^2 and x^{-1} are equal

$$\therefore -5a^4 b = 10a^3 b^2$$

$$10a^3 b^2 + 5a^4 b = 0 \checkmark$$

$$5a^3 b (2b + a) = 0$$

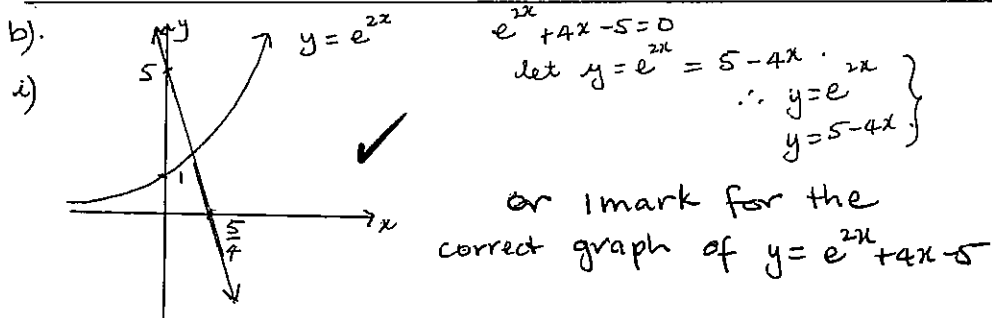
$\therefore 2b + a = 0$ as required.

(3)

Question 13.

a) no. of ways = ${}^9P_5 - {}^7P_3$
 $= 126 - 35$
 $= 91$ ✓

(2)



ii) $f(x) = e^{2x} + 4x - 5$
 $f(0) = e^0 - 5 = -4 < 0$
 $f(1) = e^2 + 4 - 5 = 6.39 > 0$
 $\therefore$ a root lies between 0 and 1 as $f(x)$ neg to pos

iii) let $x_1 = 0.5$
 $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $f(0.5) = -0.2817 \dots$
 $f'(x) = 2e^{2x} + 4$
 $f'(0.5) = 9.436 \dots$
 $= 0.5 - \frac{-0.2817}{9.436} \dots$
 $= 0.529 \dots$
 $= 0.53$ (2dp) ✓

(4)

c. i) $\ddot{x} = 2e^x$
 $\text{since } \ddot{x} = \frac{d}{dx}(\dot{x}v)$
 $\text{then } \frac{d}{dx}(\dot{x}v) = 2e^x$
 $\frac{1}{2}v^2 = 2 \int e^x dx$
 $\frac{1}{2}v^2 = 2e^x + C$
 $\text{at } v = -2, x = 0$
 $2 = 2 \times 1 + C$
 $C = 0$
 $\therefore v^2 = 4e^x$

1 mark working towards
 1 mark

(2)

Quest 13 Cont.

ii) $V^2 = 4e^x$
 $V = \pm \sqrt{4e^x}$
 $= \pm 2e^{\frac{x}{2}}$

now $V < 0$ initially
 $\therefore V = -2e^{\frac{x}{2}}$

in terms of t

$\frac{dx}{dt} = -2e^{\frac{x}{2}}$

$\frac{dt}{dx} = -\frac{1}{2}e^{-\frac{x}{2}}$

$t = e^{-\frac{x}{2}} + C$

at $t = 0, x = 0$

$\therefore 0 = e^0 + C$

$C = -1$

$\therefore t = e^{-\frac{x}{2}} - 1$

$t + 1 = e^{-\frac{x}{2}}$

$\ln(t+1) = -\frac{x}{2}$ (as $\ln e = 1$)

$\therefore x = -2 \ln(t+1)$

(3)

d) an example $-kt$

i) $V = 500 - 500e^{-kt}$

$\frac{dV}{dt} = k \cdot 500e^{-kt}$

but $\therefore 500e^{-kt} = 500 - V$

$= k(500 - V)$

$= -k(V - 500)$ as required

✓ any solid method

ii) $V = 0, t = 0, V = 21 \text{ m/sec } t = 0.958$

$V = 500 - 500e^{-5kt}$

$21 = 500 - 500e^{-5kt}$

$-479 = -500e^{-5kt}$

$0.958 = e^{-5kt}$

$\ln(0.958) = \ln e^{-5kt}$

$\therefore k = \frac{\ln 0.958}{-e}$

$= 0.008581 \dots$

$= 0.00858$ (3 sig fig)

iii) $V = ?$ when $t = 20 \text{ sec}$

$V = 500 - 500e^{-20k}$

$= 78.8546$

$\approx 78.9 \text{ m/sec}$ (1dp)

iv) as $t \rightarrow \infty, e^{-kt} = \frac{1}{e^{kt}} \rightarrow 0$

$\therefore V \rightarrow 500 \text{ m/sec}$ ✓

(4)

Question 14.

a) $x = 30\sqrt{2}t$ $y = 30\sqrt{2}t - 5t^2$

i) $x = 120\text{ m}$ $t = ?$

$120 = 30\sqrt{2}t$ height: $y = 30\sqrt{2}t - 5t^2$
 $t = \frac{4}{\sqrt{2}} \text{ sec}$ ✓
 $= 30\sqrt{2} \left(\frac{4}{\sqrt{2}}\right) - 5\left(\frac{4}{\sqrt{2}}\right)^2$
 $= 120 - 40$
 $= 80\text{ m}$ ✓

ii) horizontal range: $y = 0$ $t = ?$

$30\sqrt{2}t - 5t^2 = 0$

$5t(6\sqrt{2} - t) = 0$

$t = 0$ $t = 6\sqrt{2}$

range: $x = 30\sqrt{2} \times 6\sqrt{2}$
 $= 360\text{ m}$ ✓

Therefore this particle will travel 360m horizontally.

(3)

b) i) $P(x_1, y_1)$ $Q(x_1 + a, y_2)$ since $x^2 = 4ay$

$y = \frac{x^2}{4a}$

midpt c $\left(\frac{x_1 + x_1 + a}{2}, \frac{\frac{x_1^2}{4a} + \frac{(x_1 + a)^2}{4a}}{2}\right)$ $y_2 = \frac{(x_1 + a)^2}{4a}$ ✓

$= \left(\frac{2x_1 + a}{2}, \frac{x_1^2 + x_1^2 + 2ax_1 + a^2}{8a}\right)$ ✓

$= \left(\frac{2x_1 + a}{2}, \frac{2x_1^2 + 2ax_1 + a^2}{8a}\right)$

(2)

Quest. 14. Cont.

b ii)

$x = \frac{2x_1 + a}{2}$

$2x = 2x_1 + a$

$x_1 = \frac{2x - a}{2}$

$y = \frac{2x_1^2 + 2ax_1 + a^2}{8a}$

$y = \frac{2\left(\frac{2x - a}{2}\right)^2 + 2a\left(\frac{2x - a}{2}\right) + a^2}{8a}$ ✓

$= \frac{\cancel{2}(4x^2 - 4ax + a^2) + 2ax - \cancel{a^2} + \cancel{a^2}}{8a}$

$= \frac{4x^2 - 4ax + a^2 + 2ax - a^2 + a^2}{16a}$

$= \frac{4x^2 + a^2}{16a}$ ✓

$\therefore 16ay = 4x^2 + a^2$

$4x^2 = 16ay - a^2$

$\therefore$ a parabola (the vertex is $(0, \frac{a}{16})$) (2)

c) i) $\sum_{r=1}^n r^3 = \frac{1}{4} n^2 (n+1)^2$

test true: $\angle \text{HS} = 1^3$ $\text{RHS} = \frac{1}{4} \times 1 \times (1+1)^2$
 For $n=1$ $= 1$ $= \frac{1}{4} \times 4 = 1$

$\therefore \angle \text{HS} = \text{RHS}$ — true for $n=1$

assume true, $\sum_{r=1}^n r^3 = 1 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{4} (n^2)(n+1)^2$
 for $n=k$, $S_k = 1 + 2^2 + 3^2 + \dots + k^2 = \frac{1}{4} k^2 (k+1)^2$

prove true: ie. $S_k + T_{k+1} = S_{k+1}$
 For $n=k+1$.

$S_{k+1} = \frac{1}{4} (k+1)^2 (k+2)^2$

$$\begin{aligned} \text{LHS} &= \frac{1}{4} k^2 (k+1)^2 + (k+1)^3 \\ &= \frac{(k+1)^2}{4} (k^2 + 4(k+1)) \end{aligned}$$

must have $= \frac{1}{4} (k+1)^2 (k+2)^2$
 $= \text{RHS}$

* $\therefore$ If true for $n=k$ now proved true for $n=k+1$

Since true for $n=1$ now true for $n=1+1=2$, $n=3$
 and so on by the principles of M.I for all n .

③ marks - correct

② marks - one error ① Mark - 2 errors.

ii) $y = x^3 + x$

into $\square h = \left(\frac{1}{n}\right)^3 + \frac{1}{n}$

Area $= \frac{1}{n} \times \left(\frac{1}{n^3} + \frac{1}{n}\right)$ ✓

width $= \frac{1}{n}$

2nd strip $h = \left(\frac{2}{n}\right)^3 + \frac{2}{n}$ last strip $h = \left(\frac{n-1}{n}\right)^3 + \frac{n-1}{n}$

$A = \frac{1}{n} \left(\left(\frac{1}{n}\right)^3 + \frac{2}{n} \right)$ $A = \frac{1}{n} \times \left(\left(\frac{n-1}{n}\right)^3 + \frac{n-1}{n} \right)$
 ✓ working towards A_n .

$\therefore$ Total Area $A_n = \frac{1}{n} \times \left[\left(\frac{1}{n^3} + \frac{1}{n}\right) + \left(\left(\frac{2}{n}\right)^3 + \frac{2}{n}\right) + \left(\left(\frac{3}{n}\right)^3 + \frac{3}{n}\right) \right.$
 $\left. + \dots + \left(\left(\frac{n-1}{n}\right)^3 + \frac{n-1}{n}\right) \right]$

$\therefore$ two parts $A_n = \frac{1}{n} \sum_{r=1}^{n-1} \left(\frac{r}{n}\right)^3 + \frac{1}{n} \sum_{r=1}^{n-1} \frac{r}{n}$ ✓ splitting
 $= \frac{1}{n^4} \sum_{r=1}^{n-1} (r)^3 + \frac{1}{n^2} \sum_{r=1}^{n-1} r$

iii) Area $= \lim_{n \rightarrow \infty} A_n$ and $\sum_{r=1}^n r^3 = \frac{1}{4} (n)^2 (n+1)^2$ from (i)

$A_n = \frac{n^2 (n+1)^2}{4 n^4} + \frac{1}{n^2} \sum_{r=1}^{n-1} r$

now $\sum_{r=1}^{n-1} r$ is a series (AP)
 $1 + 2 + 3 + \dots + n-1$

$\therefore S_n = \frac{n}{2} (a+l)$
 $= \frac{n-1}{2} (1+n-1)$
 $= \frac{(n-1)}{2} (n)$

$\therefore A_n = \frac{n^2 (n+1)^2}{4 n^4} + \frac{1}{n^2} \times \frac{(n-1)}{2} n$
 $= \frac{1}{4} \cdot \frac{(n+1)^2}{n^2} + \frac{1}{2} \cdot \frac{(n-1)}{n}$

$\lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \left[\frac{1}{4} \left(1 + \frac{2}{n} + \frac{1}{n^2} \right) + \frac{1}{2} \left(1 - \frac{1}{n} \right) \right]$
 $= \frac{1}{4} + \frac{1}{2}$ ✓ as $\frac{1}{n}, \frac{1}{n^2} \rightarrow 0$ as $n \rightarrow \infty$
 $= \frac{3}{4} v^2$

* needed to show working that achieves $\frac{1}{4}$ and $\frac{1}{2}$.